

MANANK PATEL

✉ contact@manank.in ☎ +91-8320541977  [manank20](https://www.linkedin.com/in/manank20)  manank.in  github.com/manank20

EDUCATION

Birla Institute of Technology and Science - Pilani, Goa Campus	2020 - 2024
B.E. Computer Science	CGPA: 8.17/10.0

PROFESSIONAL EXPERIENCE

Software Engineer	July 2024 - Present
Arista Networks, Pune, India	

- Working on distributed, scalable backend services for large-scale wireless infrastructure, with a focus on reliability, data consistency, and secure system design
- Built and improved core state-processing and API paths, making operational data more consistent, easier to query, and easier to serve reliably at scale
- Improved the reliability of an internal device simulator by reworking its concurrency model and adding better recovery and shutdown handling for large-scale test workloads
- Worked on system stability and performance using profiling and debugging tools such as eBPF, gdb, and core dump analysis
- Rolled out sanitizer-based testing and improved crash diagnostics, making it easier to catch memory corruption, data races, and undefined behavior earlier in development

Software Engineering Intern	Jan 2024 - June 2024
Arista Networks, Pune, India	

- Improved database interactions for cloud-managed wireless services, cutting memory footprint by 6-8x and making the system more cost-efficient at scale
- Investigated scalability bottlenecks across queueing, caching, and database maintenance, and proposed a memory-safe caching redesign to improve throughput and reliability

Software Engineering Intern	May 2023 - July 2023
Arista Networks, Pune, India	

- Containerized a legacy production service and helped define its deployment lifecycle, making it easier to run, upgrade, and operate in cloud-native environments
- Supported a production Linux platform migration to improve compatibility, security, and long-term maintainability

GSoC Contributor	May 2023 - Aug 2023
OSPO UC Santa Cruz	Remote

- Built an io_uring-based communication backend for a networked key-value store to improve throughput and reduce latency
- Tuned io_uring parameters for the workload and benchmarked the design against the older synchronous implementation
- Project notes and relevant links: manank.in/blog/gsoc-blogs

RESEARCH EXPERIENCE

Undergraduate Research Student	May 2022 - Dec 2023
DaSH Lab, CSIS, BITS Goa	

- Researched and benchmarked how Linux io_uring could reduce latency in data management systems, using eBPF, perf, and flame graphs to study and improve I/O behavior.
- Studied and implemented distributed systems concepts such as Raft, Paxos, Kafka, RabbitMQ, GFS, and Hadoop, building a strong foundation in storage and cloud systems.

TECHNICAL SKILLS

Languages: C, Go, Rust, Python, SQL, Assembly, Bash

Systems Programming: POSIX signal handling, async-signal-safe code, libevent, threading models (pthreads, mutex, rwlock), memory management, reliable UDP, SSL/TLS

Performance Engineering: eBPF, perf, flamegraphs, AddressSanitizer, ThreadSanitizer, UndefinedBehaviorSanitizer, lock contention analysis, io_uring

Cloud & Data: Protocol Buffers, gRPC, distributed telemetry systems, columnar and relational databases, PostgreSQL tuning and optimization

Infrastructure: Docker, container orchestration, CI/CD pipelines, Trivy (CVE scanning), RPM packaging

Networking: Wireshark, tcpdump, network protocol analysis, authentication systems, WiFi protocols

Architecture: Scalability analysis, distributed system design, event-loop architectures, in-memory cache design